Module 07 Lab Reflective Journal

John Grubbs

I liked this assignment because it showed me that accuracy does not tell the whole story. The confusion matrix helped me see the correct and incorrect predictions in more detail.

I learned that precision focuses on how many positive predictions are correct, while recall focuses on finding as many actual positive cases as possible. Precision could matter when checking whether a building's foundation is unsafe because false alarms could lead to unnecessary repairs. Recall could matter more for medical tests because missing a real illness could be dangerous.

The cross-validation section also helped me understand why one train-test split may not always be reliable. Testing the model across five different groups gave a better idea of how well it performed overall.

Keira Burdine

During this lab, I learned how to evaluate a machine learning model beyond simply checking whether it runs. I used the dataset and a logistic regression model to predict survival. Creating a confusion matrix and classification report, then performed 5-fold cross-validation to examine how consistently the model performed.

The confusion matrix helped me understand correct and incorrect predictions. My model correctly predicted 90 non-survivors and 54 survivors, while making 35 incorrect predictions. Its test accuracy was approximately 80.45%. The classification report helped me compare precision, recall and F1 score for each class. I learned that the model was better at identifying passengers who did not survive than those who survived.

One challenge was a warning related to filling missing values in the Age column. When I changed the code, I initially left out the assignment that saves the filled values back to the dataset. This caused an error when training the model. I corrected the line, reran the cell and confirmed that the model trained successfully. This reminded me to pay close attention to small differences in Python syntax.

The 5-fold cross-validation produced an average accuracy of 78.68% and a standard deviation of 0.0094. These results showed fairly consistent performance across the folds. I now better understand why evaluating a model with multiple metrics and testing splits is important. Moving forward, I want to become more confident interpreting results and debugging code independently.

Luis Adrian

In this lab, I learned how to evaluate a Machine Learning model using more than just accuracy. I worked with the Titanic dataset again and used Logistic Regression to predict passenger survival. One of the most interesting things I learned was how to use a confusion matrix to understand the mistakes made by the model. I also learned about Precision, Recall, and F1-Score, and how each metric helps me understand the performance of a model in different situations.

Another important part of this lab was learning about Cross-Validation. At first, I thought that accuracy was enough to determine if a model was performing well, but I realized that using only one train-test split might not give the most reliable results. My model achieved 80% accuracy in the original evaluation and 78.68% average accuracy using 5-fold cross-validation. The standard deviation was 0.0094, which showed that the results were consistent across the different folds.

Overall, this lab helped me better understand how to evaluate Machine Learning models and why different evaluation methods are important. I feel more comfortable interpreting model results and understanding how to identify areas where a model can improve.

Khaleed Harper

While doing this lab I learned how model evaluation works, fraud detection, workflow, and more.